

AI Deep Learning: Recurrent Neural Networks Long Short-Term Memory (LSTM)

Thuan L Nguyen, PhD

Slide 2: Long Short-Term Memory Neural Networks (LSTM)



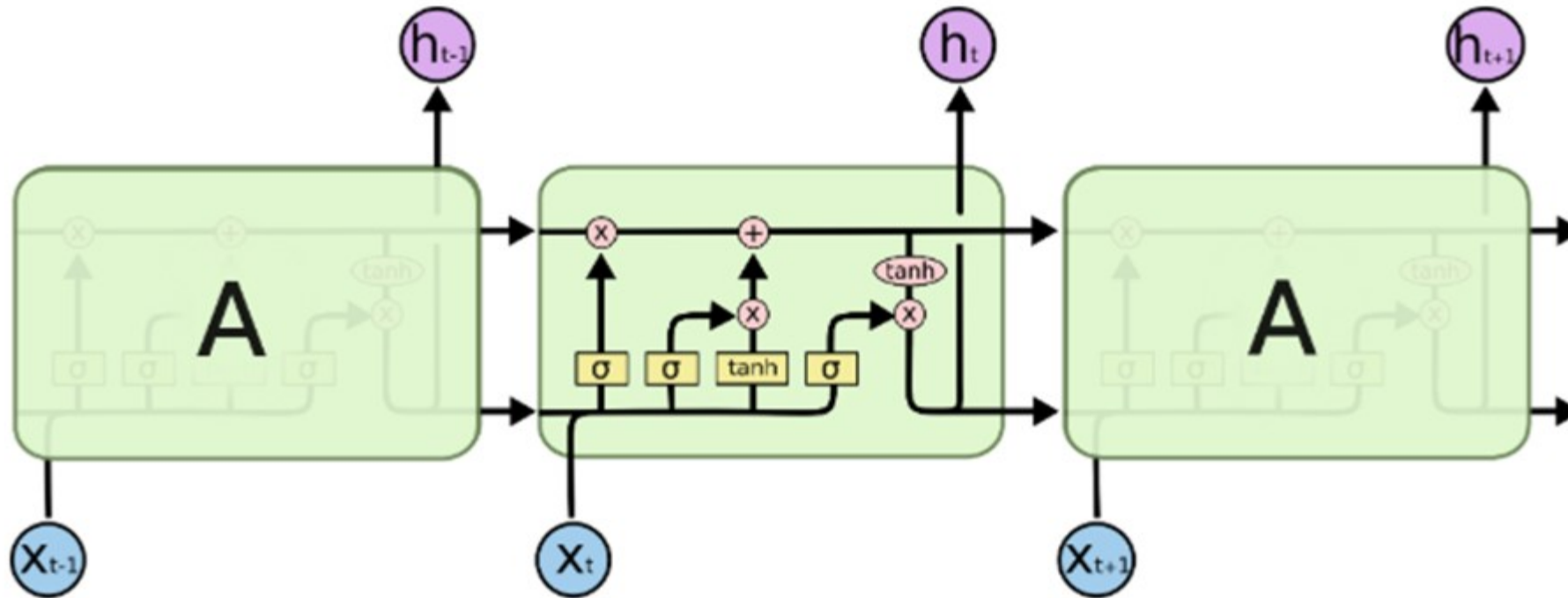
AI Deep learning (Source: mindovermachines.com)

Slide 3: Long Short-Term Memory Neural Networks (LSTM)

1. LSTM Neural Networks: Simple RNN with Short-Term Memory
2. LSTM Neural Networks : Introduction
3. LSTM Neural Networks : Core Concepts: Cell State & Gates: Overview
4. LSTM Neural Networks: Core Concepts: Cell State & Gates: Cell State
5. LSTM Neural Networks: Core Concepts: Cell State & Gates: Gates
6. LSTM Neural Networks: Flow of Information & Mathematical Models
7. LSTM Neural Networks: HOWTO Solve Vanishing Gradient Problem

Slide 4: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM: A Powerful Solution to RNN's Short-Term Memory



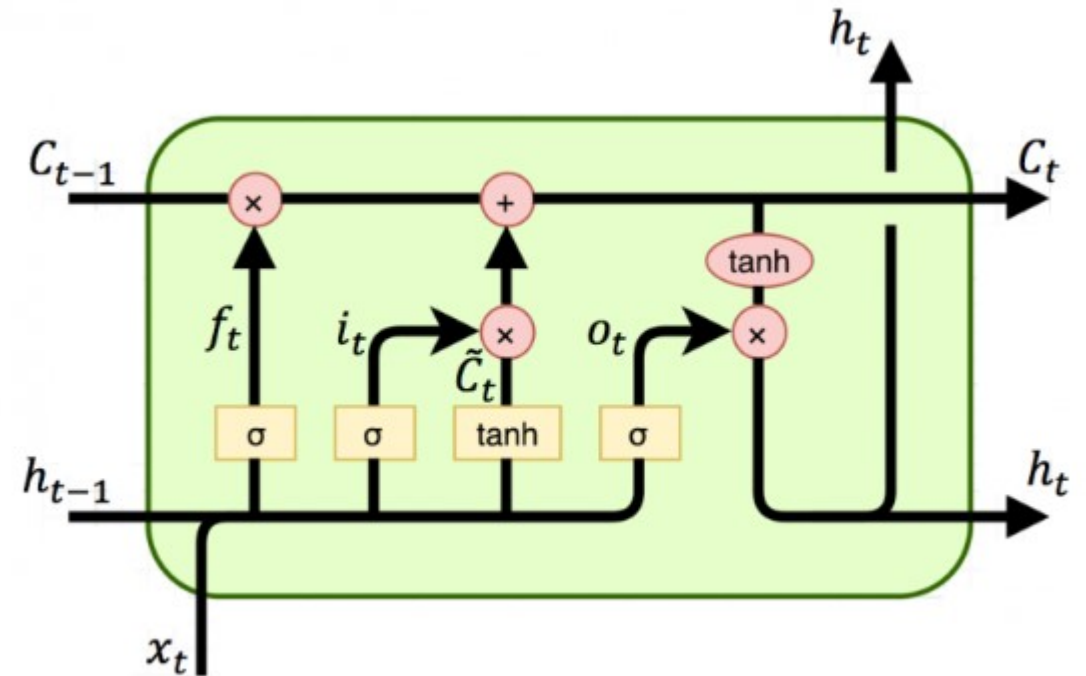
Unrolled LSTM Neural Network (Source: Colah Blogs)

Slide 5: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM: A Powerful Solution to Vanishing Gradient Problem

A Brief Introduction

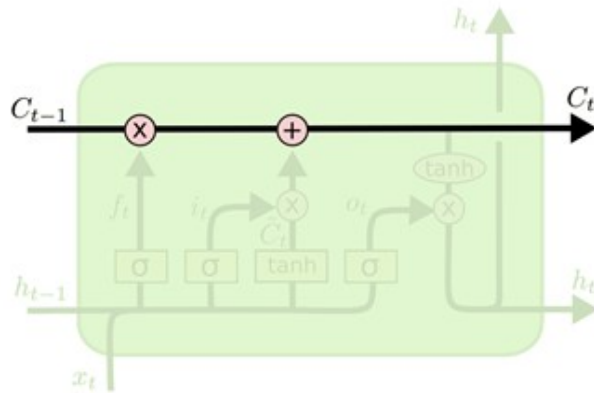
- In the mid-90s, [Sepp Hochreiter](#) and [Juergen Schmidhuber](#) (German researchers) proposed the **Long Short-Term Memory** (LSTM) neural network as a **solution** to the **vanishing gradient problem**.
- The **LSTM** network can **preserve the error** along the path of **backpropagation** through time and layers.
 - It **maintains** a **more constant error** while backpropagation is done.
 - The **neural network** can **continue learning** over many time steps.



LSTM Cell (Source: Colah's Blog)

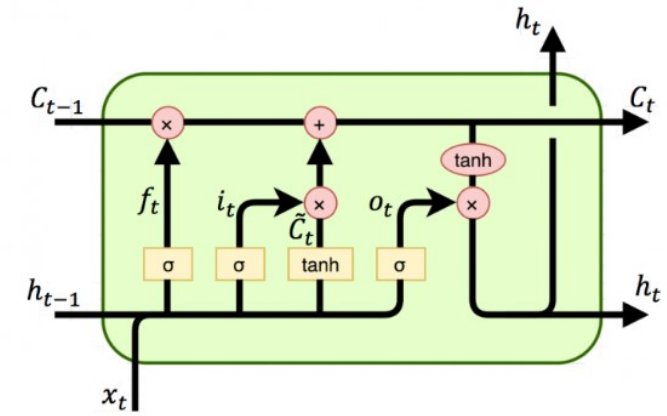
Slide 6: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM: Core Concepts



LSTM's **C State**

LSTM Cell (Source: Colah's Blog)



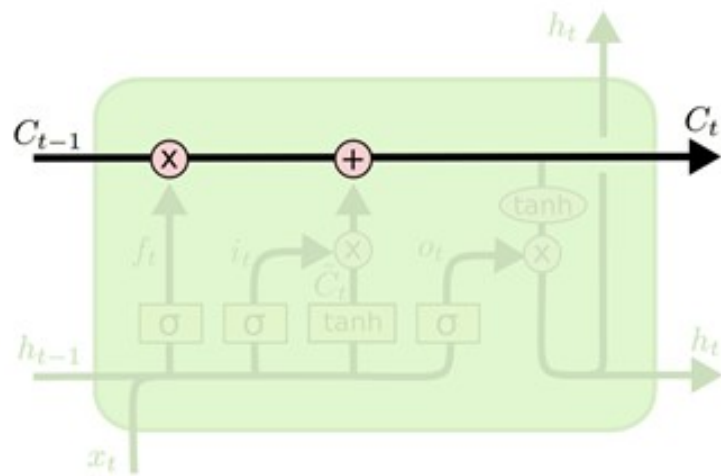
LSTM's **Gates**

LSTM: Core Concepts: Cell (C) State and Gates

- Two **most important concepts** of the LSTM neural network: **Cell state** (C state) and **gates**.
- Thanks to the Cell state and gates:
 - **LSTM can address the Simple RNN's shortcomings** caused by the short-term memory due to the vanishing gradient problem.

Slide 7: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM: Core Concepts: Cell (C) State and Gates



LSTM Cell (Source: Colah's Blog)

LSTM Cell: Cell (C) State:

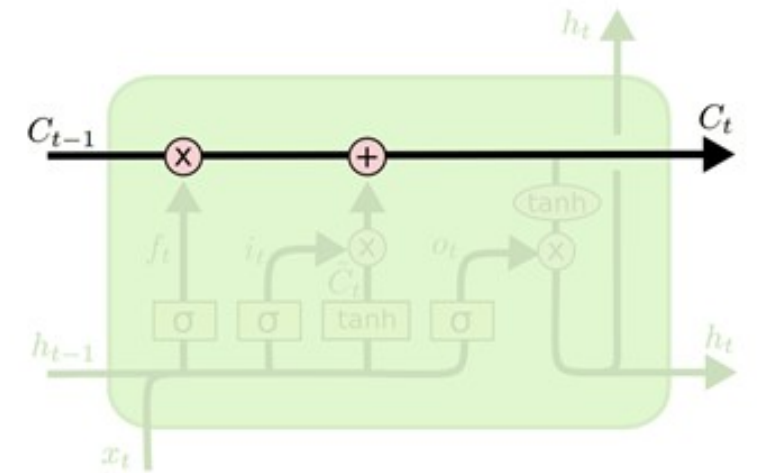
- An LSTM cell has **C state**, or **Cell state**, in addition to H state, or Hidden state.
- Any **later time step** (or layer) can use information carried in from the **previous Cell state** without any reduction, **if** the information is still **good**, i.e., **relevant** and **significant**.
- WHY?
 - Because there is **no weight** involved with the **Cell state** that can stay as it is through multiple layers or time steps.

Slide 8: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM: Core Concepts: Cell (C) State and Gates

LSTM: Cell (C) State:

- Being considered as an **information transporting** channel:
 - Able to **transfer information** from the **earliest** time steps all the way **to the end** of the **sequence chain**.
 - In theory, can **carry relevant information** throughout the **processing of the sequence**.
- Can **mitigate** the problem of short-term memory
- Being a **part of the neural network's "memory,"** in addition to the hidden (H) state.

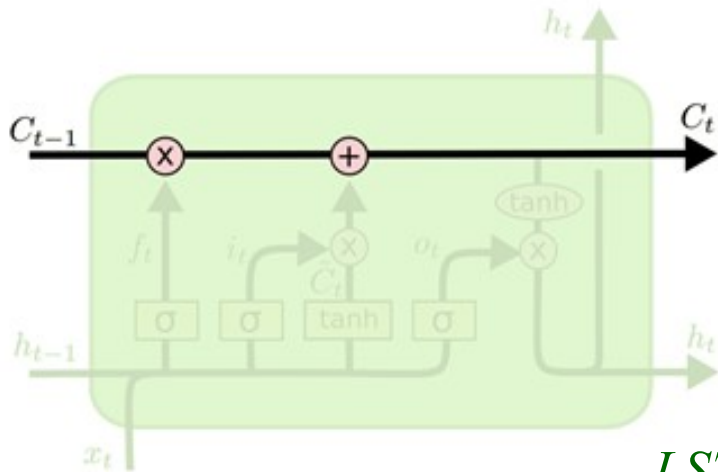


LSTM Cell (Source: Colah's Blog)

Slide 9: Long Short-Term Memory Neural Networks (LSTM)

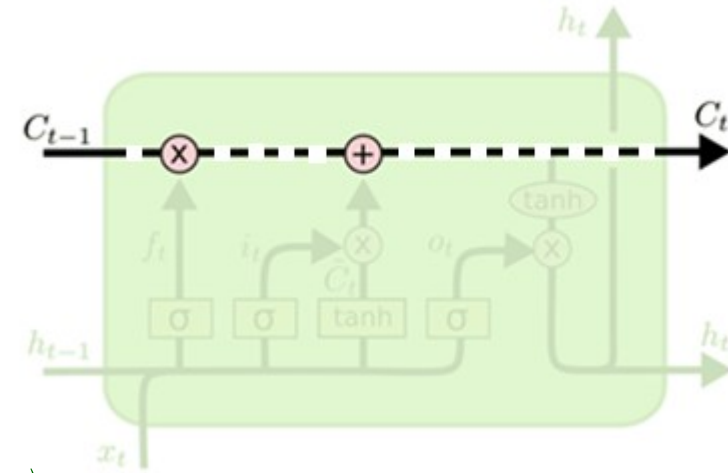
RNN: LSTM: Core Concepts: Cell (C) State and Gates: EXTRA

LSTM: Cell (C) State:



LSTM Cell (Source: Colah's Blog)

LSTM: Cell (C) State:



FLOW in a **CHANNEL**:

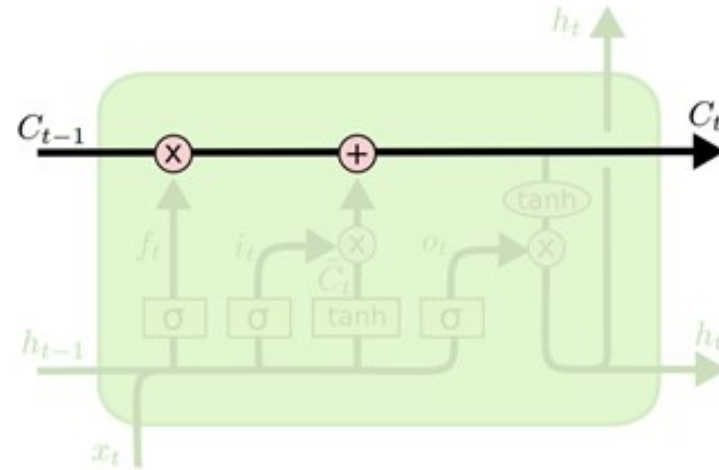
Information can **flow freely**, not being reduced throughout the channel: from the **start** until the **end** of the sequence.

SNAPSHOT of the **STATE** of a **CELL**

The **Cell (C) State** can “**memorize**” the **information** in a **cell** at **each time step**, no matter how early the information is in the sequence.

Slide 10: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM: Core Concepts: Cell (C) State and Gates



LSTM Cell (Source: Colah's Blog)

LSTM Cell: Cell (C) State:

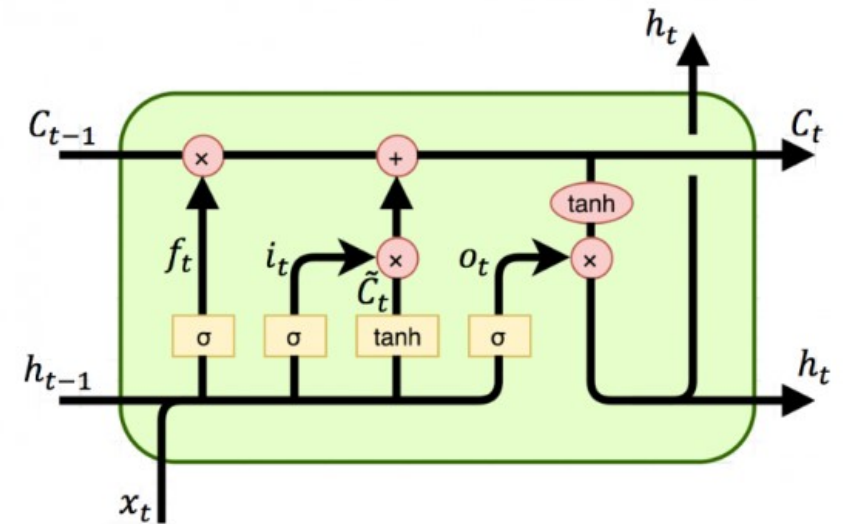
What are the differences between the **Cell (C)** State and the **Hidden (h)** state?

Slide 11: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM Cell: Cell (C) State and Gates

LSTM Cell: Gates

- **Cell state** transfers the relevant **information** along the sequence chain.
- Along this path, **information** can be **added** or **removed** to the flow, i.e., the cell state, **via gates**.
- In other words, an **LSTM cell** uses the **gates** to regulate the **flow of information** moving along the **path of processing** the sequence chain.
- There are **three gates**: **Forget**, **Input**, and **Output** in an LSTM Cell, each with one **sigmoid** activation function.



LSTM Cell (Source: Colah's Blog)

Slide 12: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM Cell: Cell (C) State and Gates

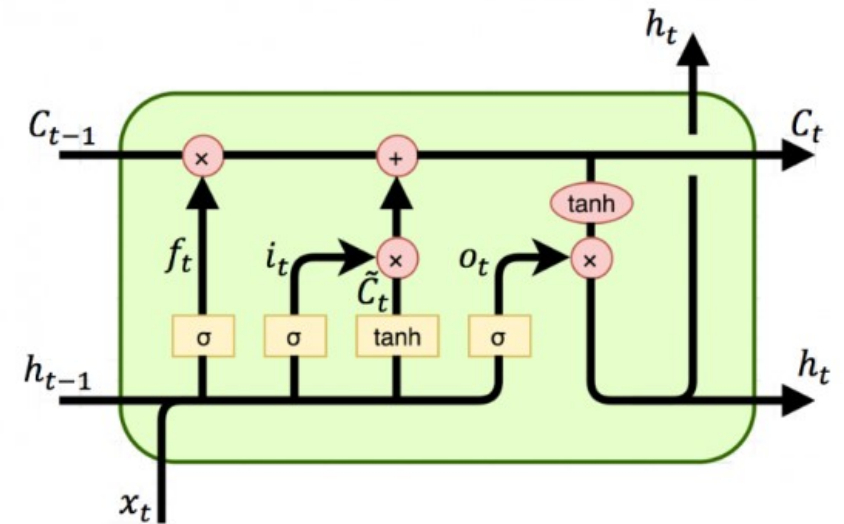
LSTM Cell: Gates

- Each **gate** is a **neural network** that can **learn** to determine what **information** should be **relevant** or **significant** and then be **used**.

NOTES:

--> Each **gate** can also be considered as a **neural network layer**.

- Each **gate** can **learn** to **decide**, during the training:
 - What **information** to **keep** because it is relevant.
 - What **information** to **“forget”** because it is irrelevant or insignificant.



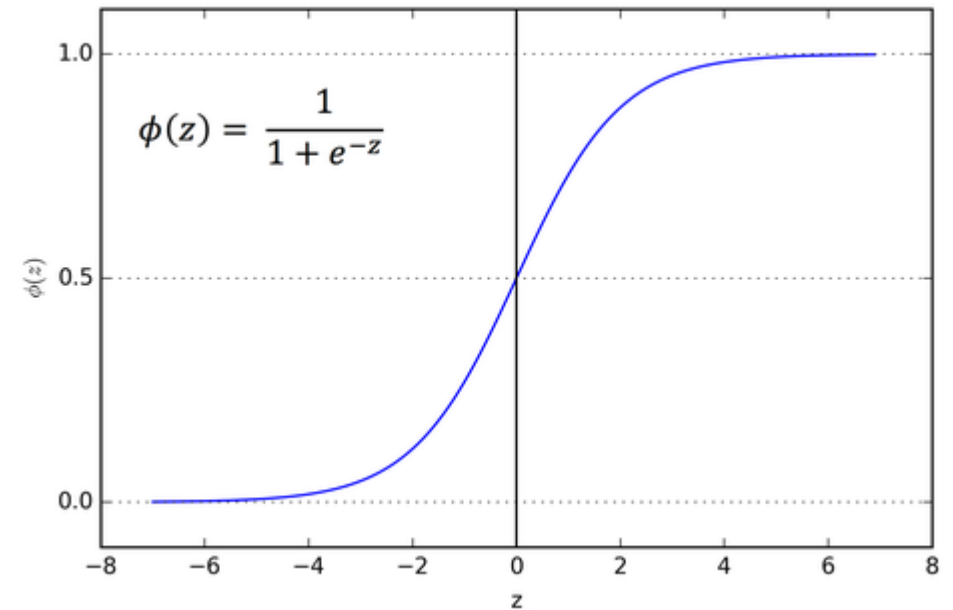
LSTM Cell (Source: Colah's Blog)

Slide 13: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM Cell: Cell (C) State and Gates

LSTM Cell: Gates: Activation Function: Sigmoid

- **Gates** contains **sigmoid** function as **activations**.
- A **sigmoid** activation has **values** between 0 and 1.
- A **sigmoid** function can be used to **keep** or “**forget**” data:
 - Any number getting **multiplied by 0** is 0.
 - → Disappearing or be “forgotten.”
 - Any number **multiplied by 1** is the same value.
 - → Staying the same or being “kept.”
- With the **sigmoid** activation function, a **neural network**:
 - Can **learn** which **data is relevant** → **keep** it
 - Can **learn** which **data is irrelevant** → “**forget**” it



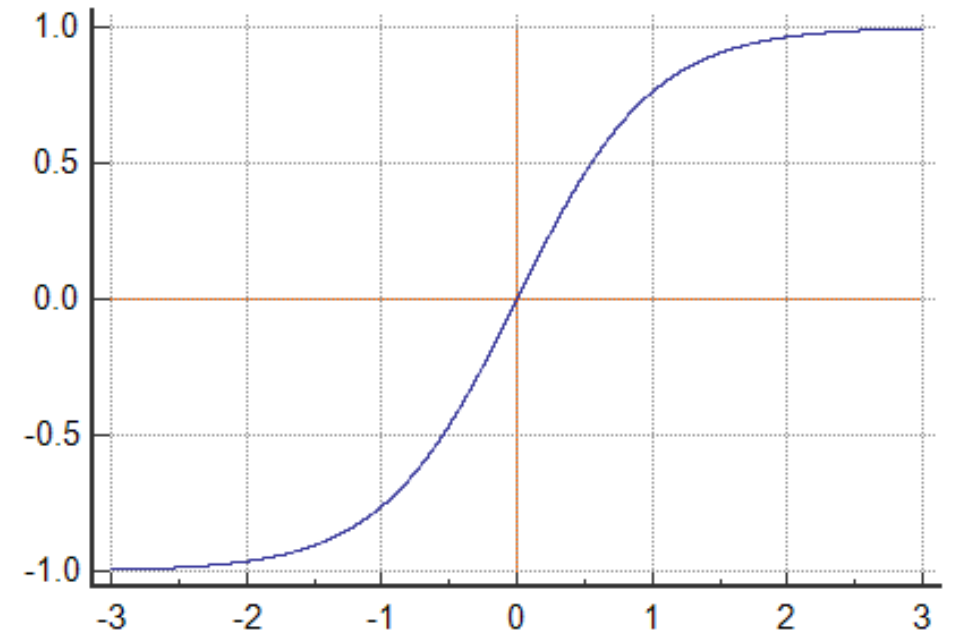
Sigmoid Function (Source: Wikipedia)

Slide 14: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM Cell: Cell (C) State and Gates

LSTM Cell: Activation Function: Tanh

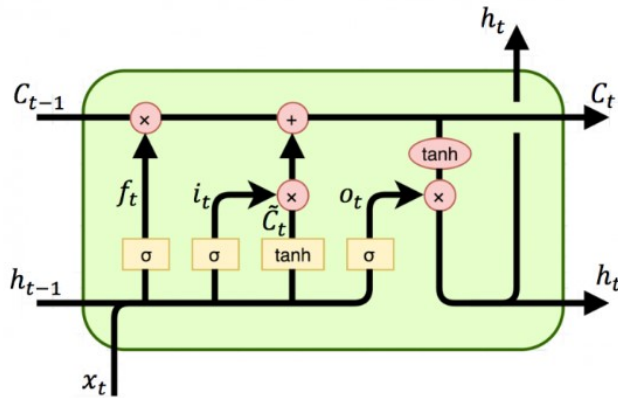
- LSTM cells also contain **tanh** function as **activations**.
- A **tanh** activation function has **values between -1 and 1**.



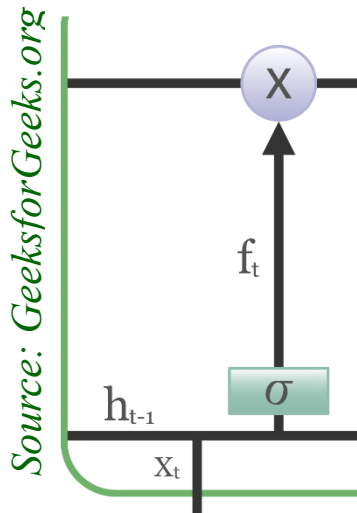
Tanh Function (Source: Medcalc.org)

Slide 15: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM Cell: Cell (C) State and Gates



LSTM Cell (Source: Colah's Blog)



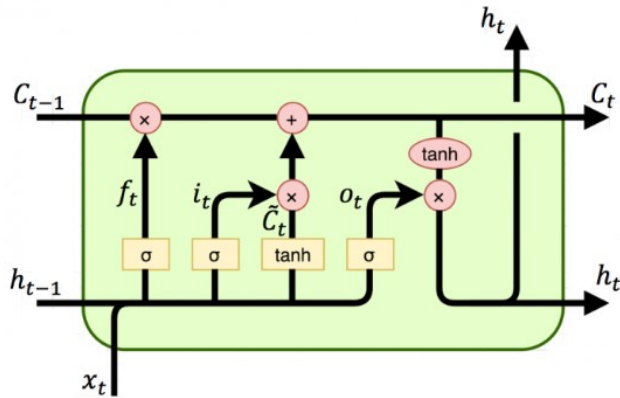
Forget Gate

LSTM: Gates: Forget Gate $\rightarrow f(t)$

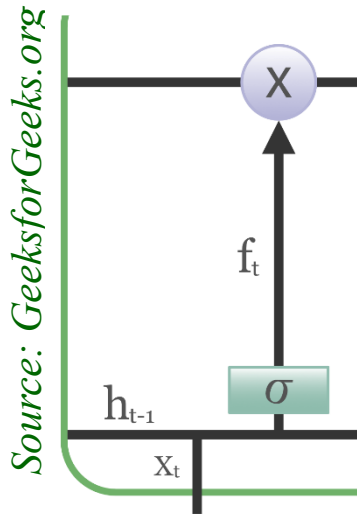
- Decide what information of the previous Cell state C_{t-1} should be **removed** (irrelevant or insignificant) or **kept** (relevant or significant).
- The output is a **vector of values** between **0** and **1**, called **Remember vector $f(t)$** , that is used to **update** the previous Cell state C_{t-1} .
- How** does the **forget gate** come up with **these values**?
 - The **gate** reviews the values of the inputs (h_{t-1} and x_t) to **determine** what information of the previous Cell state is significant (**to keep**) and what is not (**to remove** or forget).
 - The **gate** should **learn this skill** during the **training process**.

Slide 16: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM Cell: Cell (C) State and Gates



LSTM Cell (Source: Colah's Blog)



LSTM: Gates: Forget Gate $\rightarrow f(t)$

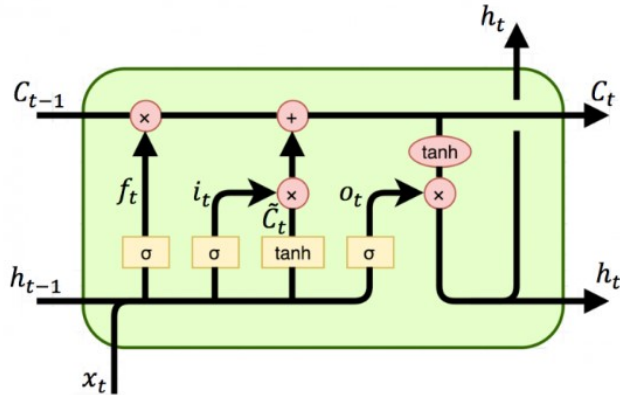
- Inputs:
 - h_{t-1} (previous hidden state) and current inputs x_t
- Outputs: $f(t) = \sigma(W_f * [h_{t-1}, x_t] + b_f)$
 - “Remember” vector
 - A vector of values between 0 and 1.
 - 1: “Completely relevant or significant”: keep 100%
 - 0: “Completely irrelevant or insignificant”: remove 100%.

NOTES:

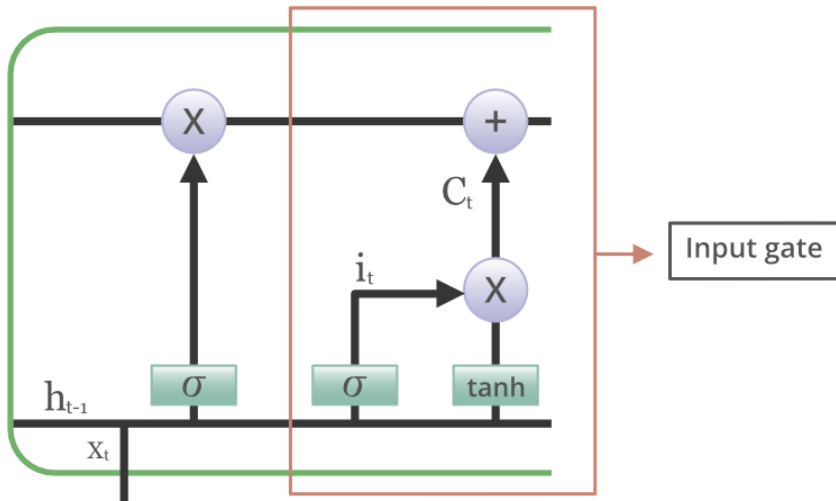
--> The output is used to **update** the previous Cell state C_{t-1} .

Slide 17: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM Cell: Cell (C) State and Gates



LSTM Cell (Source: Colah's Blog)



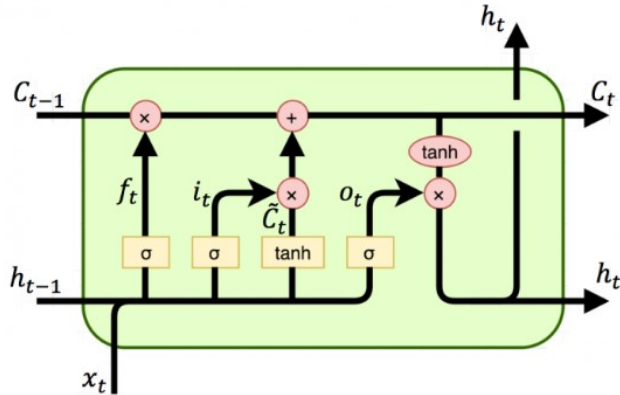
Input Gate (Source: GeeksforGeeks.org)

LSTM: Gates: **Input Gate** $\rightarrow i(t) * C(temp)$

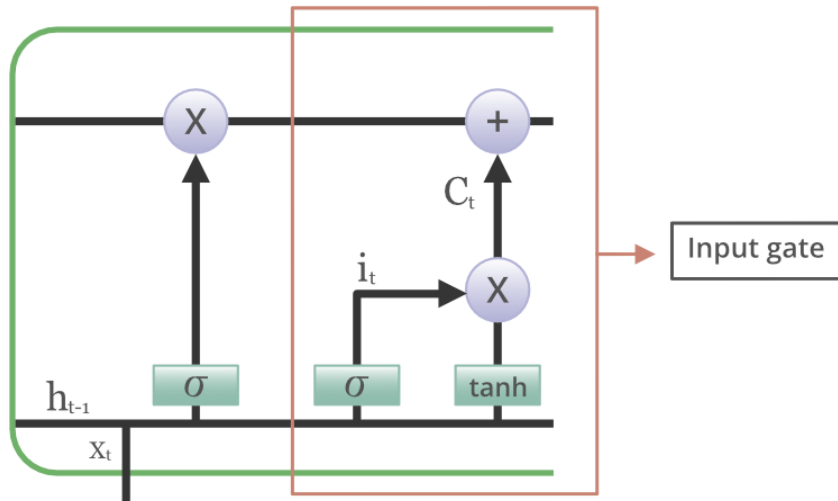
- Decide **what information** – from the inputs (h_{t-1} and x_t) – is **useful** and should be **added or saved** into the **new current Cell state $C(t)$** .
- **Modulate** the inputs (h_{t-1} and x_t) into the values between **-1** and **1**.
- The **output** is a vector of values, called the **Save vector**, representing the **useful information** from the **inputs**.
- The **output** will become **part of the new current Cell state $C(t)$** .

Slide 18: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM Cell: Cell (C) State and Gates



LSTM Cell (Source: Colah's Blog)



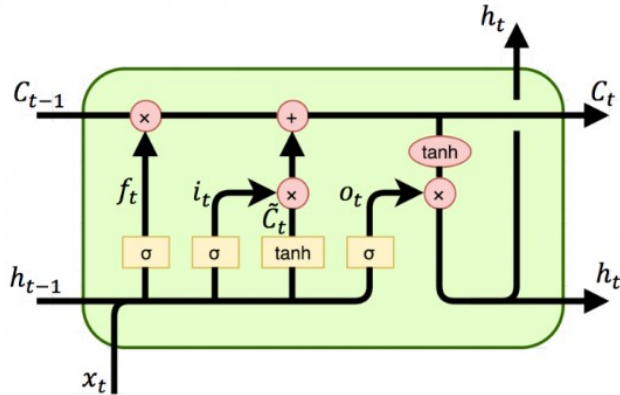
Input Gate (Source: GeeksforGeeks.org)

LSTM: Gates: Input Gate $\rightarrow i(t) * C(temp)$

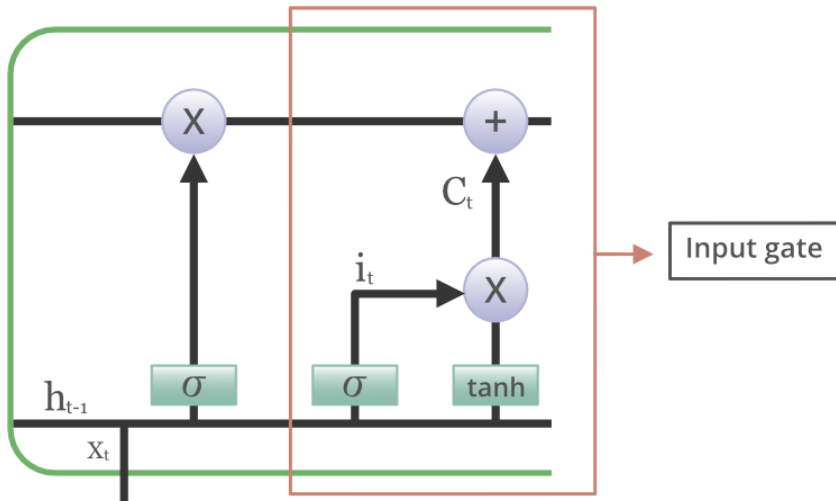
- **First**, the **inputs** (h_{t-1} and x_t) are filtered by the function **sigmoid** to be determined what should be kept.
 - Output: **$i(t)$**
- **Next**, the **inputs** (h_{t-1} and x_t) are processed by the function **tanh**, which produces a **vector of values between -1 and 1**.
 - Output: **$C(temp)$**
- **Finally**:
 - The intermediate outputs – **$i(t)$** and **$C(temp)$** – are **multiplied** to obtain the **final output** of the **Input gate**.
 - The **final output** is the useful information from the inputs that will become **part of the new Cell state $C(t)$** at the time step t .

Slide 19: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM Cell: Cell (C) State and Gates



LSTM Cell (Source: Colah's Blog)



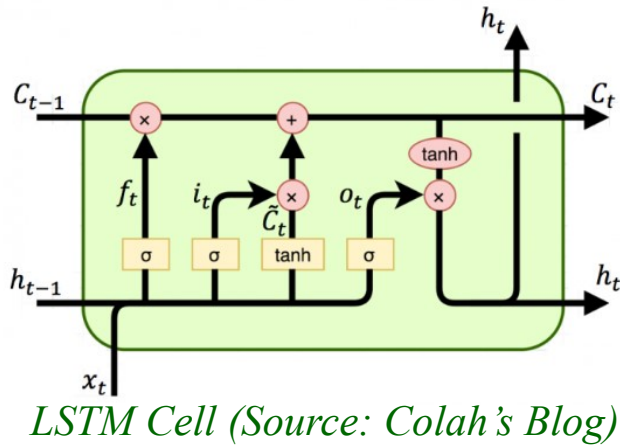
Input Gate (Source: GeeksforGeeks.org)

LSTM: Gates: Input Gate $\rightarrow i(t) * C(temp)$

- Inputs:
 - h_{t-1} (previous hidden state) and current inputs x_t
- Intermediate Output 1: $i(t) = \sigma(W_i * [h_{t-1}, x_t] + b_i)$
 - A vector of values between 0 and 1.
 - 1: "Completely relevant or significant": keep 100%
 - 0: "Completely irrelevant or insignificant": remove 100%.
- Intermediate Output 2: $C(temp) = \tanh(W_c * [h_{t-1}, x_t] + b_c)$
 - A vector of values between -1 and 1.
- Output: $i(t) * C(temp) \rightarrow$ Save vector
 - The final output is the useful information from the inputs that will become part of the new Cell state $C(t)$.

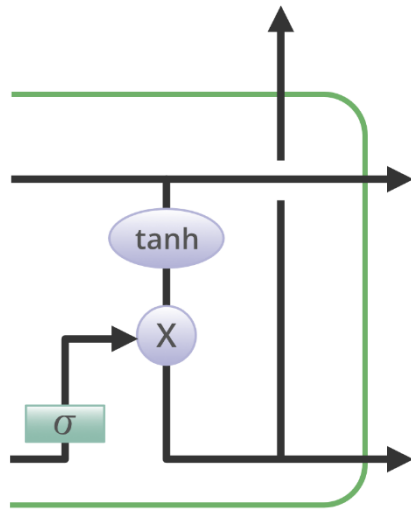
Slide 20: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM Cell: Cell (C) State and Gates



LSTM: Gates: Output Gate $\rightarrow C(t)$ and $h(t)$

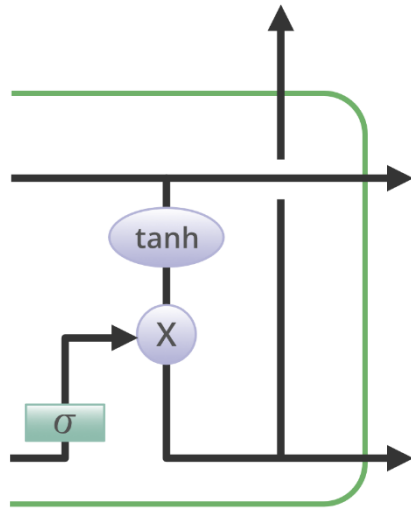
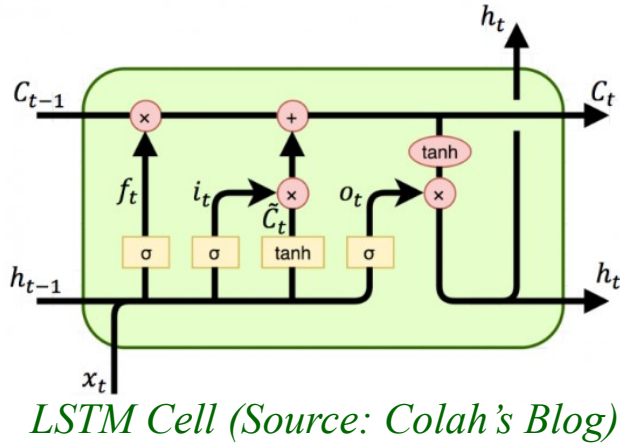
- Compute and extract the new current Cell state $C(t)$ at the time step t .
- Decide **what information** – from the inputs (h_{t-1} and x_t) and the new current Cell state $C(t)$ – should be used to make the new current hidden state $h(t)$ at the time step t .
- The **output** should be a vector of values, called the **focus vector**, representing the new current hidden state $h(t)$ of the cell at the time step t .



Output Gate (Source: GeeksforGeeks.org)

Slide 21: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM Cell: Cell (C) State and Gates



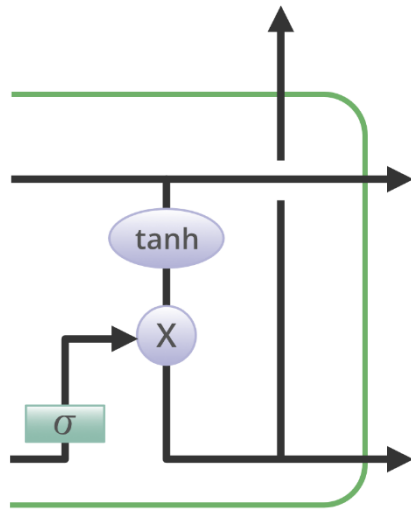
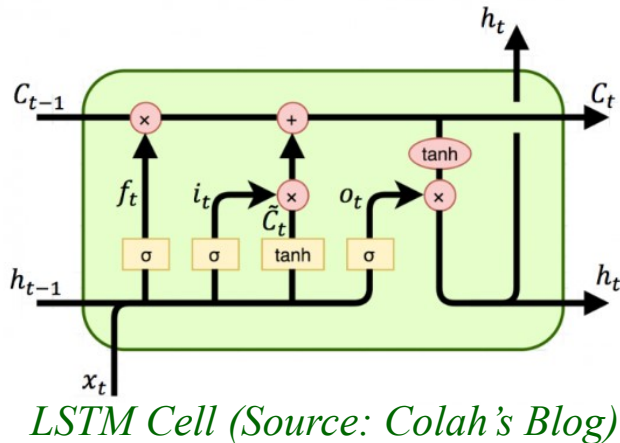
Output Gate (Source: GeeksforGeeks.org)

LSTM: Gates: Output Gate $\rightarrow C(t)$

- Compute and extract the new current Cell state $C(t)$ at the time step t .
- Inputs:
 - “Remembered” information from the previous Cell state C_{t-1} .
 - = Output of Forget gate: $f(t) * C_{t-1}$
 - Useful or relevant information from the inputs (h_{t-1} and x_t)
 - = Output of Input gate: $i_t * C(temp)$
- Output: $C(t) = f_t * C_{t-1} + i_t * C(temp)$
 - The new current Cell state $C(t)$ at the time step t .

Slide 22: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM Cell: Cell (C) State and Gates



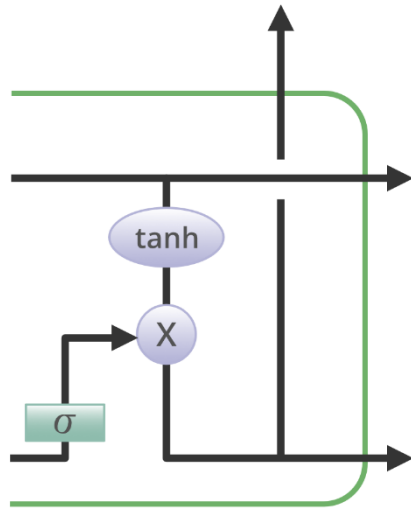
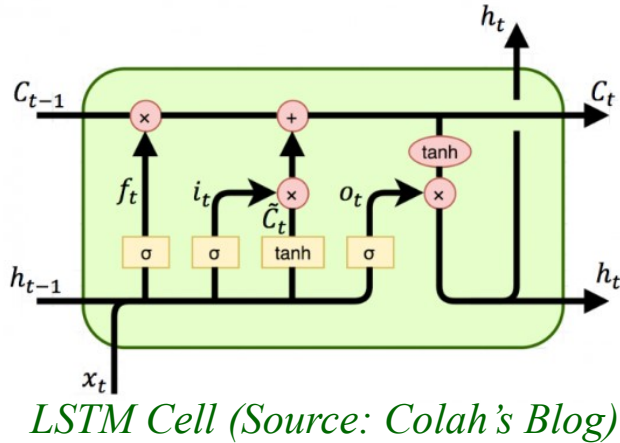
Output Gate (Source: GeeksforGeeks.org)

LSTM: Gates: **Output Gate** $\rightarrow h(t)$

- **First**, the inputs (h_{t-1} and x_t) are filtered by the function **sigmoid** to be determined **what should be kept**, which produces a **vector of values between 0 and 1**.
 - Output: **$o(t)$**
- **Next**, the **new current Cell state $C(t)$** is processed by the function **tanh**, which produces a **vector of values between -1 and 1**.
 - Output: **$\tanh(C_t)$**
- **Finally**:
 - The intermediate outputs – **$o(t)$** and : **$\tanh(C_t)$** – are **multiplied** to obtain the **final output** of the **Output gate**.
 - The **final output** is the **new Hidden state $h(t)$** at the **time step t** .

Slide 23: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM Cell: Cell (C) State and Gates



LSTM: Gates: **Output Gate** $\rightarrow h(t)$

- Inputs:
 - h_{t-1} (previous hidden state) and current inputs x_t
 - The new current Cell state $C(t)$ at the time step t .
- Intermediate Output 1: $o(t) = \sigma(W_o * [h_{t-1}, x_t] + b_o)$
 - A vector of values between 0 and 1.
 - 1: “Completely relevant or significant”: keep 100%
 - 0: “Completely irrelevant or insignificant”: remove 100%.
- Intermediate Output 2: $\tanh(C_t)$
 - A vector of values between -1 and 1.
- Output: $h(t) = o(t) * \tanh(C_t) \rightarrow$ **Focus** vector
 - The final output is the new Hidden state $h(t)$.

Slide 24: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM Cell: Cell (C) State and Gates

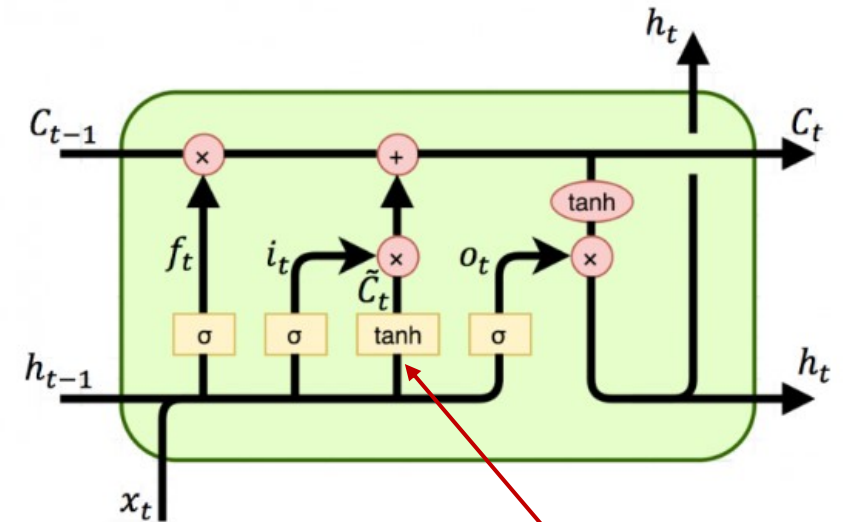
LSTM Cell: Gates

NOTES:

--> Some authors suggest that there is the 4th gate in an LSTM cell: The input modulation gate.

--> The opinion is fine for personal preference.

--> It is widely accepted that the LSTM cell contains three main gates (forget, input, and output) that are matched with three sigmoid activation functions.

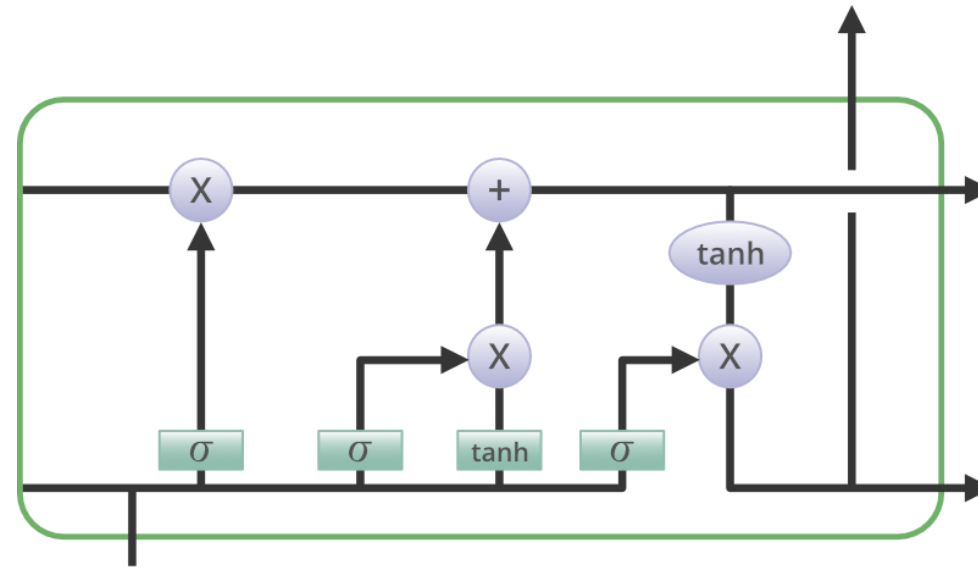


LSTM Cell (Source: Colah's Blog)

Input Modulation Gate

Slide 25: Long Short-Term Memory Neural Networks (LSTM)

RNN: LSTM Cell: Cell (C) State and Gates



LSTM Cell: Gates (Source: GeeksforGeeks.org)

LSTM Cell: Gates

Why can each **gate** be a neural **network**?

Why can each **gate** be also a neural network **layer**?